Finance 372T/397 – Group Project

Overview

The final project is a group-based applied project in which you design, build, and evaluate an AI-powered tool or analysis relevant to investment management. Your project must add value for a clearly defined investment decision and demonstrate how AI improves analysis, scale, or decision-making while preserving human judgment.

The project is worth **25 points** (25% of your course grade).

Suggested Project Ideas

You have wide latitude to choose a project. Here are some examples to get you started:

1. Invent and backtest a novel AI-driven quant strategy
2. Comprehensive textual analysis platform (e.g., earnings calls, news, filings)
3. Quant analysis platform (input ticker → output metrics and recommendations)
4. Retirement savings planner with AI-driven assumptions
5. Custom AI agent for investment research
6. Analyst report peer reviewer / quality checker
7. Social media sentiment tracker for market signals

These are suggestions, not an exhaustive list. The best projects tend to solve a problem you personally find interesting.

Rules

- Groups of at most **4** students. Groups with students in different sections are allowed provided the full group is able to attend the same session during their presentation.
- No collaboration across groups. You may use textbooks, data sources, your computer, your notes, and AI tools, but each group must do its own work.
- You must pre-commit to a division of labor plan in your pitch. Grades may differ across group members if the group reports that someone fails to complete their assigned tasks.
- You are encouraged to use AI tools for every stage of the project. You are solely responsible for the quality of your submitted work.

Schedule and Deadlines

Deliverable	Date	Time	Points
Group Selection	Thursday 4/2	11:59pm (Canvas)	2
Pitch (slides)	Tuesday 4/7	12:00pm (Canvas)	5
Presentation (slides)	Tue 4/21 or Thu 4/23	12:00pm (Canvas)	8
Final Writeup	Monday 4/27	11:59pm (Canvas)	10
		Total	**25**

Grading Rubric

1. Group Selection (2 points)

Thursday 4/2 at 11:59pm

Submit your group roster on Canvas, full credit for on-time submission.

2. Pitch (5 points)

Tuesday 4/7 in class — slides due on Canvas by 12:00pm

Each group submits **3 slides** on Canvas by noon and presents the first two in class:

1. The idea: what you plan to build and why it is useful
2. Execution plan: how you will build it and the timeline
3. Division of labor (not presented): who is responsible for what

You have a maximum of **4 minutes** to present slides 1–2, followed by **3 minutes** for suggestions and Q&A from the class and me.

Criterion	Description
Usefulness	Does the project address a real investment need? Would someone actually use this?
Novelty	Is the idea creative or differentiated, or is it something trivially available already?
Feasibility	Can this group realistically execute this project in three weeks?
Execution plan	Is there a clear, specific plan for who does what and when?
Presentation polish	Clear delivery, good use of time, well-organized slides. See the Presentation Style Guide below.

The pitch is your chance to get early feedback. Come with a concrete idea, not a vague direction.

3. Presentation (8 points)

Tuesday 4/21 & Thursday 4/23, slides due on Canvas by 12:00pm on your presentation day

Groups will be **randomly assigned** to present on one of the two days. Each group has **7 minutes** to present, followed by **4 minutes** of Q&A, with 2 minutes between groups for transitions. Submit your slides on Canvas by noon on your assigned presentation day.

Criterion	Description
Minimum viable product	You must have a working product or completed analysis to show, not just plans or mockups.
Explanation: what & why	Clearly explain what your project does and the investment problem it solves.
Explanation: how	Walk through your methodology, data, and AI tools used.
Presentation polish	Well-organized slides, clear delivery, good use of time, smooth transitions between speakers. See the Presentation Style Guide below.
Project quality	Depth and rigor of the underlying work – does it reflect serious effort and thought?

4. Final Writeup (10 points)

Monday 4/27, due on Canvas by 11:59pm

Submit your final writeup on Canvas. This is your opportunity to incorporate feedback from the presentations and polish your final product.

Criterion	Description
Final execution	The finished product should reflect improvements based on presentation feedback. Show the final, complete version of your project.
Responsiveness to feedback	Did you address the suggestions and questions raised during your presentation?
Writing quality	Clear, professional writing following the style guide below.
Project quality	Overall depth, rigor, and sophistication of the work.

Presentation Style Guide

These guidelines apply to both the pitch and the final presentation.

- **Practice getting your talk under the required time.** You have 4 minutes for the pitch and 7 minutes for the presentation. These limits feel generous until you are on stage – they go faster than you think. Time yourselves in advance and cut material rather than rushing.
- **No walls of text.** Slides are not a script. Use short bullet points, key phrases, and visuals. If your audience is reading a paragraph, they are not listening to you.
- **Everything must be legible.** Font size, axis labels, legends, table entries – if someone in the back row cannot read it, it should not be on the slide. When in doubt, make it bigger.
- **Figures and diagrams are good.** A well-chosen chart, screenshot, or workflow diagram communicates more than a page of text. Use visuals to show results, architecture, or key examples.
- **Dense tables are bad.** If you must show a table, highlight the one or two numbers that matter. Do not paste a full regression output and expect the audience to find the interesting part themselves.
- **Smooth transitions between speakers.** If multiple group members present, plan who says what and practice the handoffs. Awkward pauses and "uh, I think you're next" waste your limited time.

Writeup Style Guide

General Style and Format

- Include a **title**, followed by the students' names and an **abstract of 100 words or less** that summarizes the entire paper. If your boss is lazy and only reads the abstract, they should know (at a high level) what your project does and the key results. The paper is not a mystery novel – say up front what you find.
- Write for a hypothetical audience of **professional investors** who have strong general knowledge of finance and quant methods but need to be explained the specifics of your project. Do not cite lectures from the course – your audience did not take the course.
- Write in the **present tense** even when describing data collection or what comes later in the paper. So "we analyze" not "we analyzed" or "we will analyze."
- Do not include personal stories about how hard the project was or why you personally are interested in the topic. Your audience does not care. The motivation section should give a generally-applicable reason why your project is useful or interesting.
- Use **1.5 or double spacing**, **11 or 12pt font**. Maximum **10 pages** including everything (title, abstract, figures, tables, bibliography, etc.).

- Avoid long paragraphs or run-on sentences. You should have at least one and usually two or three paragraph breaks on every page.

Bibliography

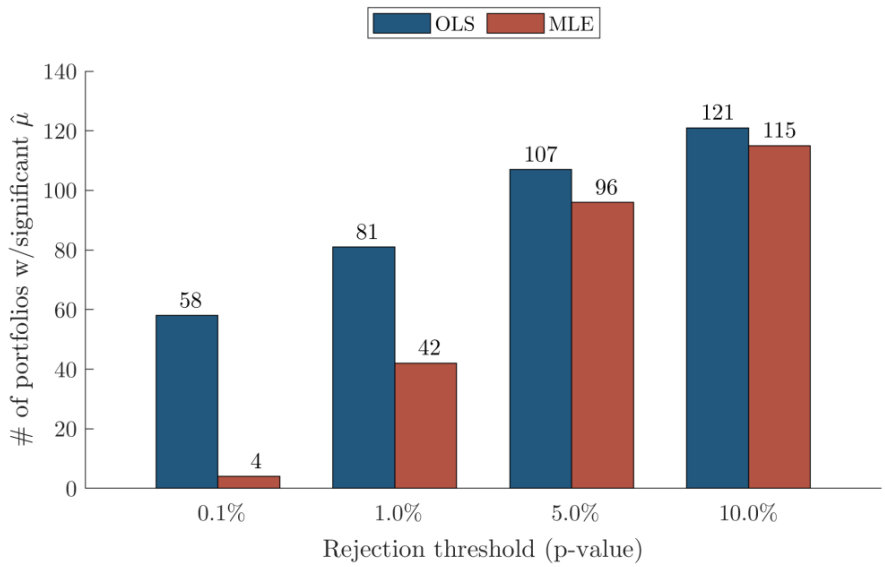
- Include a bibliography at the end following MLA format. You can get correctly formatted citations by clicking the "Cite" button on Google Scholar and copying from there.
- In the text, cite using the format: LastName1, LastName2, and LastName3 (Year). For example, "Johnson and So (2012)" or "Bybee, Kelly, Manela, and Xiu (2024)."

Tables and Figures

- Every table and figure must have a **numbered header** with a brief title and a detailed description of what is shown. See the example below. The description should be clear enough that a reader skimming through the paper can understand the exhibit without reading the surrounding text.
- Make sure every word, number, and line is legible – not too small, not too big.
- If you report regression output, include standard errors, t-statistics, or p-values.
- **Do not use too many digits.** Round to 0, 1, 2, or 3 decimal places depending on what is informative. "1.234586921234" is distracting and conveys nothing beyond "1.235." Meanwhile, "0.002" would be useless if rounded to "0.00." Use judgment.

Figure 4: Unconditional Expected Return Inferences for 174 Portfolios

This figure reports the number of characteristic-sorted portfolios for which the null hypothesis that the unconditional expected return μ is zero is rejected at different p-value thresholds. The blue bars use OLS with White (1980) standard errors. The red bars use likelihood ratio tests based on the maximum likelihood estimates of our statistical model, as described in Section 3.2.1. We conduct these tests for quarterly returns of 174 long-short equity portfolios, which include the four portfolios that we focus on in our main analysis and 170 portfolios analyzed by Chen and Zimmermann (2021). The sample period varies across portfolios and consists of at most 234 quarterly observations from Q3 1963 through Q4 2021.



Example of a well-formatted figure with a numbered title, detailed description, and clean presentation. Your figures should follow this format.